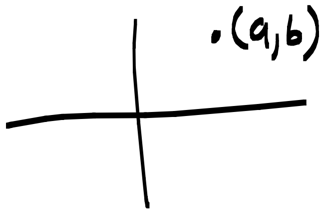
Theme 6: Vector Algebra

Unit 6.1 Three dimensional coordinate system. p 786

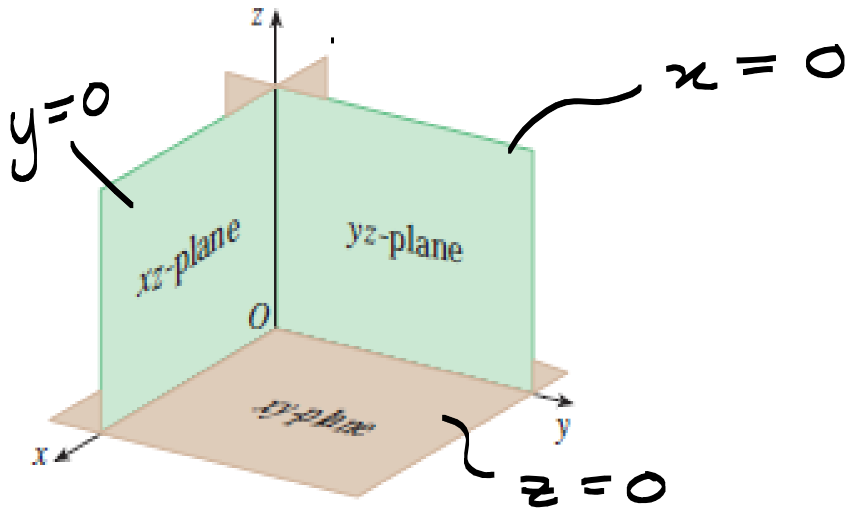
Notation $\mathbb{R}^n$

- One dimensional space $(\mathbb{R}^1)$
  a line – with point a on the line
- Two dimensional space $(\mathbb{R}^2)$
  a plane – with point (a,b)

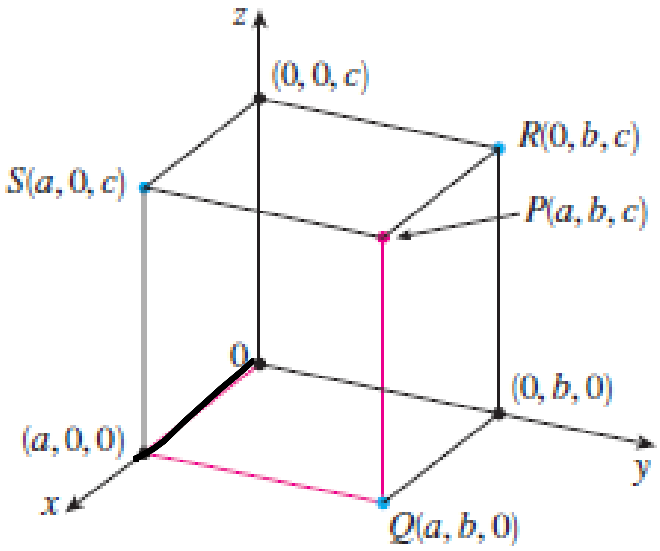


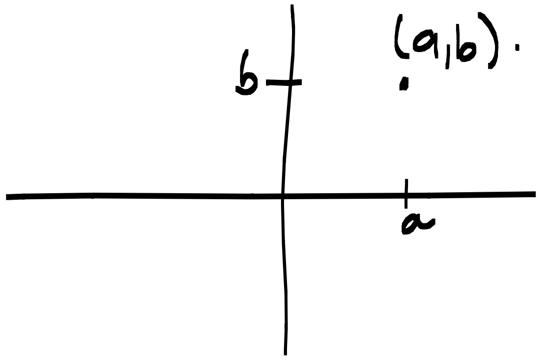
- Three dimensional space $(\underline{\underline{\mathbb{R}^3}})$
  space – with point (a, b, c)

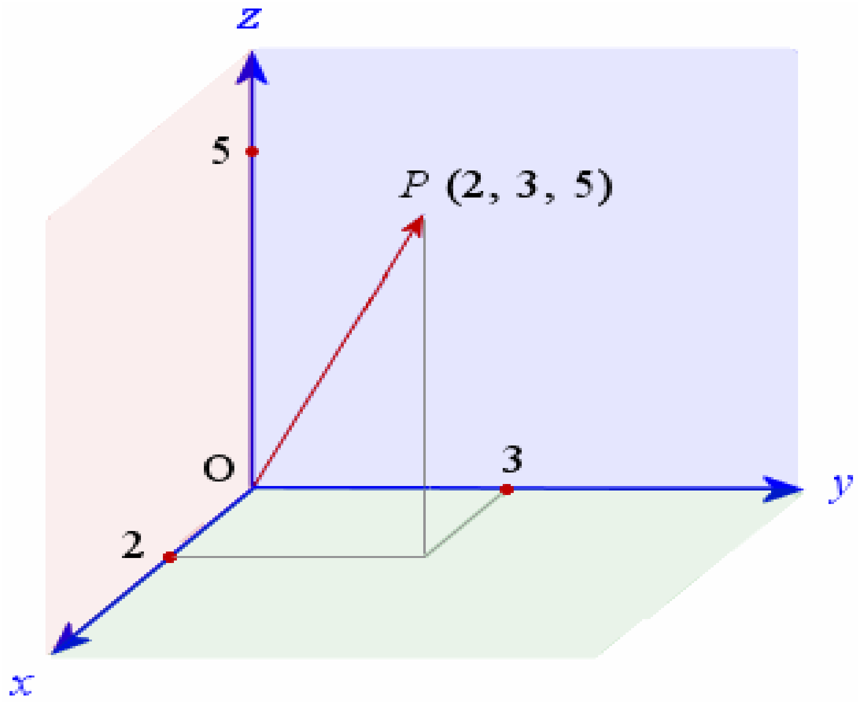
$$\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) / x, y, z \in \mathbb{R}\}$$



(a) Coordinate planes







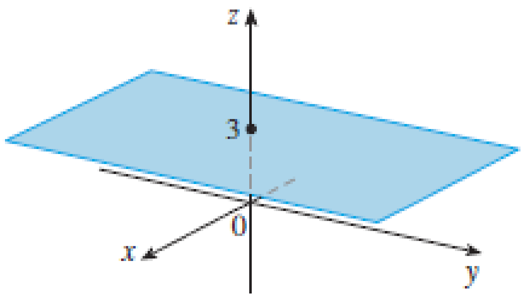
$P(2,3,5)$.

What surfaces in $\mathbb{R}^3$ are represented by:

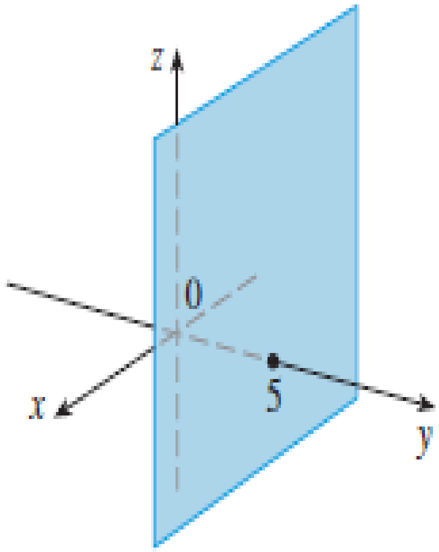
(a) $z = 3$

(b) $y = 5$

(c) $y = x$



(a) $z = 3$, a plane in $\mathbb{R}^3$



(b) $y = 5$, a plane in $\mathbb{R}^3$

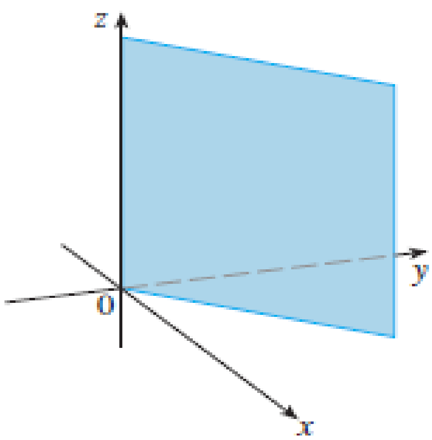
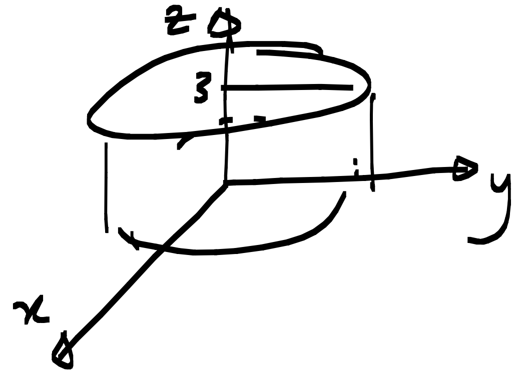
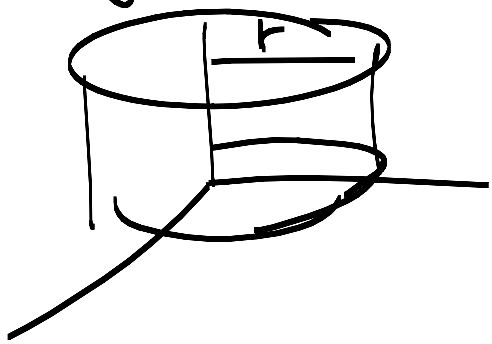


FIGURE 10
The plane $y = x$

What does $x^2+y^2=1$ and $z=3$ look like?



What does $x^2+y^2=1$ represent in $\mathbb{R}^3$?

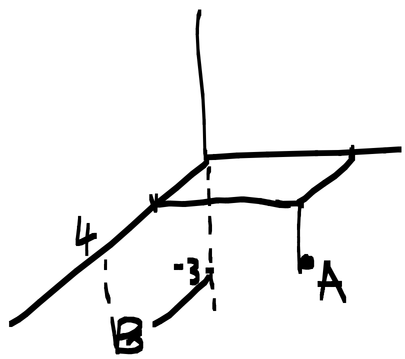


Distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ p788

$$|P_1 P_2| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

Example:

Find the distance between $A(1, 2, -3)$ and $B(4, 0, -3)$



$$|AB| = \sqrt{3^2 + 2^2 + 0^2}$$
$$= \sqrt{13}$$

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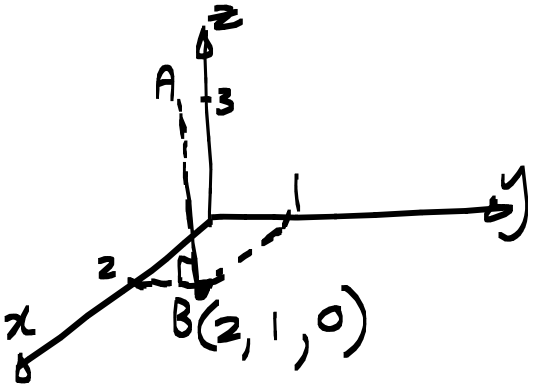
② Distance between $A(2,1,3)$ and the xy-plane.



$$|AB| = \sqrt{0^2 + 0^2 + 3^2}$$
$$= 3$$

Equation of sphere: $x^2 + y^2 = r^2$

$$(x-a)^2 + (y-b)^2 = r^2$$

is a circle in $\mathbb{R}^2$ with (centre (a,b) and radius r.

Equation of a sphere p 789

Radius r and center $C(h, k, l)$

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

Example

Find the centre and radius of the sphere in $\mathbb{R}^3$

$$x^2 + y^2 + z^2 + 2x - 6z + 8 = 0$$

Complete the square!

$$\underbrace{x^2 + 2x} + y^2 + \underbrace{z^2 - 6z} = -8$$

$$(x+1)^2 + y^2 + (z-3)^2 = -8 + 1 + 9$$

Centre $(-1, 0, 3)$ and radius $= \sqrt{2}$